If $E_j^2 \leq \frac{\varepsilon^2}{c^2(M+1)} \int_{x_{j-1}}^{x_j}$ move to next element; else refine.

→ **CG(2) method:** $\begin{cases} -(au')' = f \text{ on } (0,1) \\ u(0) = u(1) = 0 \end{cases}$

$$\int_0^1 a u' v' dx = \int_0^1 f v dx \quad \forall v \in V$$

$$V = \left\{ v : \int_0^1 (v^2 + (v')^2) dx < \infty \;,\; v(0) = v(1) = 0 \right\}$$

For CG(2):

$$V_h = \left\{ v \in V : v|_{[x_{j-1}, x_j]} = a_j + b_j x + c_j x^2 \right\}$$

Key: need to find a suitable basis for V_h.

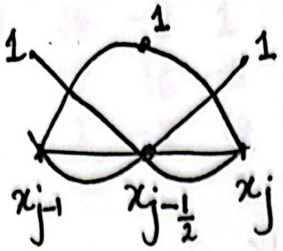
Mesh: T_h: $x_0 = 0 \quad x_1 \quad x_2 \quad x_3 \quad x_4 \cdots x_M \quad x_{M+1} = 1$

Simplest way to write CG(2) is to use Lagrangian polys. to do this, introduce a "submesh".

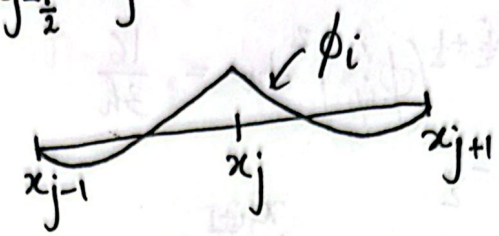
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CG(2): $x_0 \; x_{1/2} \; x_1 \; x_{3/2} \; x_2 \; x_{5/2} \; x_3 \; x_{7/2} \; x_4 \cdots x_m \; x_{M+1/2} \; x_{M+1}$

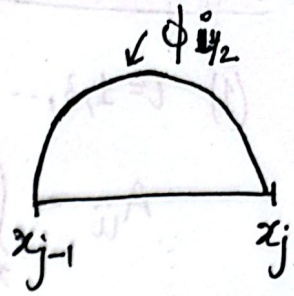
For a given element, there are 3 Lagrange polys:



2-basis function:



&



$$\phi_i(x) = \begin{cases} \dfrac{2(x-x_{i+\frac{1}{2}})(x-x_{i+1})}{(x_{i+1}-x_i)^2} & x_i \le x \le x_{i+1} \\ \dfrac{2(x-x_{i-\frac{1}{2}})(x-x_{i-1})}{(x_i-x_{i-1})^2} & x_{i-1} \le x \le x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_{i-\frac{1}{2}}(x) = \begin{cases} \dfrac{4(x_i-x)(x-x_{i-1})}{(x_i-x_{i-1})^2} & x_{i-1} \le x \le x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore V_h = \text{Span}\left\{\phi_{\frac{1}{2}}, \phi_1, \phi_{\frac{3}{2}}, \phi_2, \cdots, \phi_M, \phi_{M+\frac{1}{2}}\right\}$$

$$\therefore \text{if } U \in V_h \text{ then } U(x) = \sum_{j=1}^{2M+1} \zeta_j \phi_{j/2}(x)$$

Note: for any function $v \in V$: $\pi_h v = \sum_{K=1}^{2M+1} v(x_{\frac{K}{2}})\phi_{\frac{K}{2}}(x)$

test functions: $v = \phi_{i/2}(x)$ for $i = 1, 2, \ldots, 2M+1$.

⇒ CG(2) Var Prob:

$$\sum_{i=1}^{2M+1} \zeta_j \int_0^1 a(x)\, \phi'_{i/2} \phi'_{j/2}\, dx = \int_0^1 f \phi_{i/2}\, dx$$

:: $(2M+1)\times(2M+1)$ symmetric pos. definite system, but not tridiagonal.

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ex: consider $a \equiv 1$ & $h_j = h \ \forall j$

Stiffness matrix:

(1) $i = 1, 3, \ldots, 2M+1$

$$A_{ii} = \int_{x_{\frac{i}{2}-\frac{1}{2}}}^{x_{\frac{i}{2}+\frac{1}{2}}} \left(\phi'_{i/2}\right)^2 dx = \frac{16}{3h}$$

$$A_{(i-1,i)} = A_{(i,i-1)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \phi'_{i/2}\, \phi'_{\frac{i-1}{2}}\, dx = \frac{-8}{3h}$$

(2) $i = 2, 4, 6, \ldots, 2M$

$$A_{ii} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \left[\phi'_{i/2}\right]^2 dx = \frac{14}{3h}$$

$$A_{(i,i-1)} = A_{(i-1,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i}{2}}} \phi'_{i/2}\, \phi'_{\frac{i}{2}-\frac{1}{2}}\, dx = \frac{-8}{3h}$$

$$A_{(i,i-2)} = A_{(i-2,i)} = \int_{x_{\frac{i-1}{2}}}^{x_{i/2}} \phi'_{i/2}\, \phi'_{\frac{i}{2}-1}\, dx = \frac{1}{3h}$$

RHS vector:

$$f \approx \tilde{\Pi}_h f = \sum_{j=0}^{2M+2} f(x_{j/2})\, \phi_{j/2}(x)$$

(1) $i = 1, 3, \ldots, 2M+1$

$$f_i = \sum_{j=1}^{2M+1} f(x_{j/2}) \int_{x_{\frac{i-1}{2}}}^{x_{\frac{i+1}{2}}} \phi_{j/2}\, \phi_{i/2}\, dx = \frac{h}{15}\left[f(x_{\frac{i-1}{2}}) + 8 f(x_{i/2}) + f(x_{\frac{i+1}{2}})\right]$$

(2) $i = 2, 4, \ldots, 2M$

$$f_i = \frac{h}{30}\left[-f(x_{\frac{i-2}{2}}) + 2f(x_{\frac{i-1}{2}}) + 8f(x_{i/2}) + 2f(x_{\frac{i+1}{2}}) - f(x_{\frac{i+2}{2}})\right]$$

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$$\therefore \underline{\underline{A}} = \frac{1}{3h}\begin{bmatrix} 16 & -8 & & & & & & \\ -8 & 14 & -8 & 1 & & & & \\ & -8 & 16 & -8 & & & & \\ & 1 & -8 & 14 & -8 & 1 & & \\ & & \ddots & \ddots & \ddots & \ddots & \ddots & \\ & & & & 1 & -8 & 14 & -8 \\ & & & & & & -8 & 16 \end{bmatrix}$$

$$\underline{\underline{A}}\,\underline{x} = \underline{b}$$

Note: energy norm error:

$$\|u-U\|_E \le \|u-v\|_E \quad \forall\, v \in V_h$$

take $v = \Pi_h u$

$$\therefore \boxed{\|u-U\|_E \le c\|h^2 u'''\|.}$$

Note: L^2 error

$$\boxed{\|u-U\| \le c\|h^3 u'''\|.}$$

⇒ Beam equation:

$$\begin{cases} u^{(iv)} = f \quad \text{on} \quad 0<x<1 \\ u(0) = u'(0) = u(1) = u'(1) = 0 \end{cases}$$

Variation Problem:

$$\int_0^1 v u^{(iv)}\,dx = \int_0^1 f v\,dx$$

$$= -\int_0^1 v' u'''\,dx + [v u''']_0^1 = \int_0^1 f v\,dx$$

let $v(0) = v(1) = 0$

$$\int_0^1 v'' u''\,dx - [v' u'']_0^1 = \int_0^1 f v\,dx$$

Let $v'(0) = v'(1) = 0$

$\therefore \int_0^1 u''v''dx = \int_0^1 fvdx$

$V = \{ v : \int_0^1 [v^2 + (v')^2 + (v'')^2] dx < \infty \quad v(0) = v(1) = v'(0) = v'(1) = 0 \}$

functions in V are $C^1[0,1]$.

• {Find $u \in V$ s.t $\int_0^1 u''v''dx = \int_0^1 fvdx \quad \forall v \in V$.} Variational problem.

⇒ Energy Minimization:

Define: $F(w) = \frac{1}{2}\int_0^1 (w'')^2 dx - \int_0^1 fwdx$

(Min Prob): Find ~~s.t~~ $u \in V$ s.t, $F(u) \leq F(w) \quad \forall w \in V$.

Claim: (Var Prob) ⇔ (Min Prob)

(⇒) Assume $u \in V$ satisfies Var. Prob:

Let $w = u + v \in V$

$F(w) = F(u+v) = \frac{1}{2}\int_0^1 (u'')^2 dx + \frac{1}{2}\int_0^1 (v'')^2 dx + \int_0^1 u''v''dx - \int_0^1 fudx - \int_0^1 fvdx$

$F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2 dx$

$\therefore F(w) \geq F(u)$

Note: $F(w) = F(u) \Rightarrow v'' \equiv 0 \Rightarrow v \equiv 0 \Rightarrow u \equiv w.$ (due to B.Cs)

⇒ CG(3) Method: $V_h \subseteq V$.

On each element: $[x_{j-1}, x_j]$

- Need a poly that matches value & derivative to its neighbors at x_{j-1} & x_j ⇒ 4 conditions.

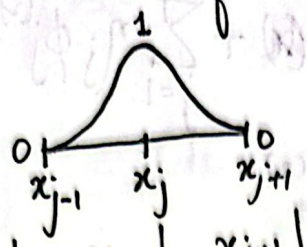
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∴ need a cubic poly (4 coefficients).

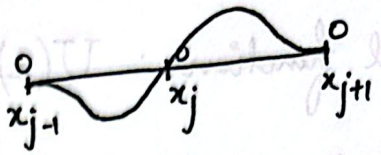
$$V_h = \{ v \in V : v|_{[x_{j-1}, x_j]} = a_j + b_j x + c_j x^2 + d_j x^3 \}$$

Key: Find a basis for V_h.

$\phi_j^{(i)}(x)$:



$\phi_j^{(2)}(x)$:



	x_{j-1}	x_j	x_{j+1}
$\phi_j^{(1)}$:	0	1	0
$(\phi_j^{(1)})'$:	0	6	0
$\phi_j^{(2)}$:	0	0	0
$(\phi_j^{(2)})'$:	0	1	0

$$\phi_j^{(1)} = \begin{cases} 3\left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^2 - 2\left(\frac{x-x_{j-1}}{x_j-x_{j-1}}\right)^3 & x_{j-1} \le x \le x_j \\ 3\left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^2 - 2\left(\frac{x-x_{j+1}}{x_j-x_{j+1}}\right)^3 & x_j \le x \le x_{j+1} \\ 0 \end{cases}$$

$$\phi_j^{(2)} = \begin{cases} -\frac{1}{h_j}(x-x_{j-1})^2 + \frac{1}{h_j^2}(x-x_{j-1})^3 & x_{j-1} \le x \le x_j \\ \frac{1}{h_{j+1}}(x-x_{j+1})^2 + \frac{1}{h_{j+1}^2}(x-x_{j+1})^3 & x_j \le x \le x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

CG(3) FEM :

Find $U \in V_h$ s.t $\int_0^1 U'' v'' dx = \int_0^1 f v dx \quad \forall v \in V_h$

$V_h = \text{Span}\{\phi_1^{(1)}, \phi_2^{(1)}, \dots, \phi_M^{(1)}, \phi_1^{(2)}, \phi_2^{(2)}, \dots, \phi_M^{(2)}\}$

$\dim V_h = 2M$

trial functions: $U(x) = \sum_{j=1}^{M} \xi_j \phi_j^{(1)}(x) + \sum_{j=1}^{M} \eta_j \phi_j^{(2)}(x)$

test functions: $v = \phi_1^{(1)}, \dots, \phi_M^{(2)}$

Also, for any $u \in V_h$:

$$\pi_h u = \sum_{j=1}^{M} u(x_j)\phi_j^{(1)}(x) + \sum_{j=1}^{M} u'(x_j)\phi_j^{(2)}(x) \quad \text{where } \pi_h u \in V_h$$

⇒ Interpolation errors:

$$\|u - \pi_h u\|_{L^2(0,1)} \le c_1 \|h^4 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \pi_h u)'\|_{L^2(0,1)} \le c_2 \|h^3 u^{(4)}\|_{L^2(0,1)}$$

$$\|(u - \pi_h u)''\|_{L^2(0,1)} \le c_3 \|h^2 u^{(4)}\|_{L^2(0,1)}$$

ex: (uniform mesh) $h_j \equiv h \ \forall j$

$$\int_0^1 U'' v'' dx = \int_0^1 f v dx$$

$$v = \phi_1^{(1)}, \phi_2^{(1)}, \dots, \phi_M^{(1)}, \phi_1^{(2)}, \dots, \phi_M^{(2)}$$

Stiffness matrix: $\int_0^1 U'' v'' dx = \sum_{j=1}^{M} \xi_j \int_0^1 (\phi_j^{(1)})''(\phi_k^{(1)})'' dx + \sum \eta_j \int_0^1 \phi_j^{(2)''} \phi_k^{1''} dx$

$k = 1, 2, \dots, M.$

$$\& \sum_{j=1}^{M} \xi_j \int_0^1 (\phi_j^{(1)})''(\phi_k^{(2)})'' dx + \sum_{j=1}^{M} \eta_j \int_0^1 (\phi_j^{(2)})''(\phi_k^{(2)})'' dx$$

$k = 1, 2, \dots M$

$$\begin{bmatrix} \underline{\underline{A}} & \vdots & \underline{\underline{C}} \\ \cdots & \cdots & \cdots \\ -\underline{\underline{C}} & \vdots & \underline{\underline{B}} \end{bmatrix} \in \mathbb{R}^{2M \times 2M}$$

$$\underline{\underline{A}}, \underline{\underline{B}}, \underline{\underline{C}} \in \mathbb{R}^{M \times M}$$

$$\rightarrow A_{jk} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(1)})'' dx$$

$$\underline{\underline{A}} = \frac{12}{h^3} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal & symmetric

$$\rightarrow B_{jk} = \int_0^1 (\phi_j^{(2)})'' (\phi_k^{(2)})'' dx$$

$$\underline{\underline{B}} = \frac{2}{h} \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & 4 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal, symmetric

$$\rightarrow C_{jk} = \int_0^1 (\phi_j^{(1)})'' (\phi_k^{(2)})'' dx$$

$$\underline{\underline{C}} = \frac{6}{h^2} \begin{bmatrix} 0 & 1 & & & \\ -1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & -1 & 0 \end{bmatrix}$$

$\therefore$ M×M, tridiagonal, $\underline{\underline{C}}^T = -\underline{\underline{C}}$

$\Rightarrow$ is anti-symmetric

$$\therefore \underline{\underline{S}} = \begin{bmatrix} \underline{\underline{A}} & \vdots & \underline{\underline{C}} \\ \cdots & \cdots & \cdots \\ \underline{\underline{C}}^T & \vdots & \underline{\underline{B}} \end{bmatrix}$$

$$\therefore \underline{\underline{S}}^T = \underline{\underline{S}}$$

$\therefore$ $\underline{\underline{S}}$ is 2M×2M, symmetric

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RHS:

$$f \to \tilde{\Pi}_h f = \sum_{j=0}^{M+1} f(x_j)\,\phi_j^{(1)}(x) + \sum_{j=0}^{M+1} f'(x_j)\,\phi_j^{(2)}(x)$$

$$\int_0^1 f v\,dx \to \int_0^1 (\tilde{\Pi}_h f)\cdot v\,dx$$

$$f_j^{(1)} = \frac{h}{70}\left[9f(x_{j-1}) + 52f(x_j) + 9f(x_{j+1})\right] + \frac{13h^2}{420}\left[f'(x_{j-1}) - f'(x_{j+1})\right]$$

$$j = 1, 2, \cdots, M$$

$$f_j^{(2)} = \frac{13h^2}{420}\left[f(x_{j+1}) - f(x_{j-1})\right] - \frac{h^3}{420}\left[3f'(x_{j-1}) - 8f'(x_j) + 3f'(x_{j+1})\right]$$

$$j = 1, 2, \cdots, M$$

∴ final 2M×2M system:

$$\begin{bmatrix} \underline{\underline{A}} & \vdots & \underline{\underline{C}} \\ \cdots & \cdots & \cdots \\ -\underline{\underline{C}} & \vdots & \underline{\underline{B}} \end{bmatrix} \begin{bmatrix} \underline{x} \\ \cdots \\ \underline{y} \end{bmatrix} = \begin{bmatrix} \underline{f}^{(1)} \\ \cdots \\ \underline{f}^{(2)} \end{bmatrix}$$

2M×2M, Symmetric, Positive definite.